R E P O R T

Need Assessment for a Desk Companion Alarm with Task Reminders

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Analysis of the Survey Results

Introduction

To evaluate the relevance of the proposed desk companion, a survey was conducted. The results reveal two main observations: students regularly struggle to remember important tasks, and the digital tools they currently use do not always meet their needs. This analysis focuses on these two aspects.

Methodology

The survey was created with Google Forms and sent to every student at IMT Atlantique between 21 September and 5 October. It received 131 responses. The questionnaire combined multiple-choice questions and open-ended questions, covering how often students forget tasks, the tools they use, how effective phone notifications are, their interest in a physical reminder, the features they want and the price they would pay.

The answers were classified into two themes: (1) forgetting tasks and (2) current tools and expectations. For each multiple-choice question, we retrieved the percentage of respondents per answer and compared related questions, for example how often students ignore phone notifications against their interest in a physical device. The results were then summarised into key takeaways and turned into design decisions for the product.

1. Forgetting Tasks: A Common Student Problem

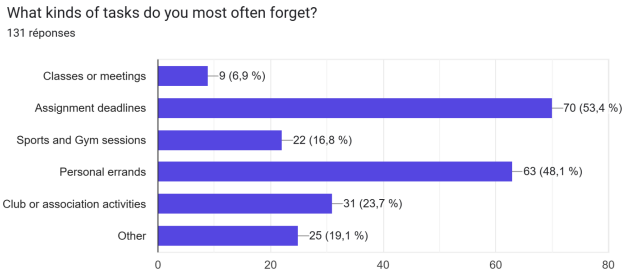
The survey shows that forgetting tasks is a frequent issue among students. More than 40% of respondents reported that they sometimes or often forget important responsibilities, indicating that organization remains a challenge despite the many tools available today.



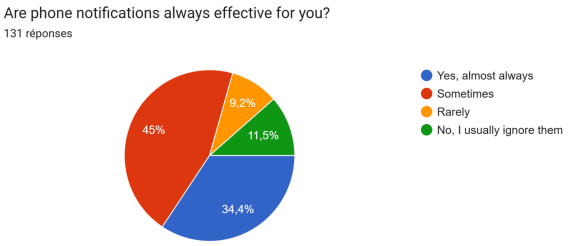
> *Key Takeaway: Over 40% of students regularly struggle with task management, highlighting a clear need for improved reminder mechanisms.*

The most commonly forgotten items are assignment deadlines (53.4%), followed by personal errands (48.1%). This is particularly significant because missed academic deadlines can directly impact students' performance and increase stress levels.

> *Critical Impact: Academic assignments are the primary source of forgotten deadlines, directly impacting student performance.*



Another important finding is the phenomenon of notification fatigue. Many students admitted that they have ignored reminders on their phones and later forgotten the associated task. Since students receive a constant flow of notifications throughout the day, important reminders can easily be overlooked.



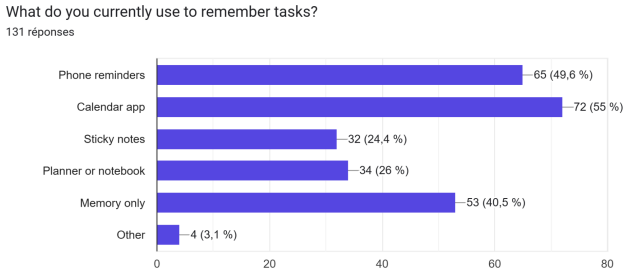
> *Notification Fatigue: Mobile alerts lose effectiveness due to constant digital noise, leaving users prone to ignoring crucial reminders.*

Comparing these charts shows that the problem does not come from a lack of tools: 42% of students forget important tasks at least sometimes, while only 34.4% consider phone notifications almost always effective.

Overall, the survey confirms that remembering deadlines and important activities remains a real issue for a large proportion of students, creating an opportunity for alternative reminder solutions.

2. Current Tools: Useful but Not Fully Effective

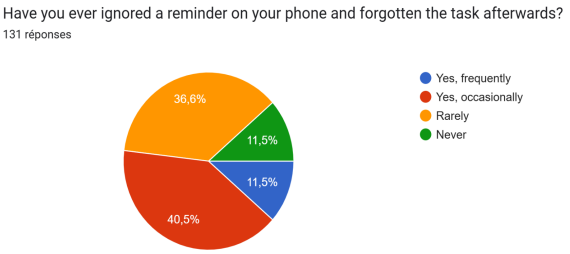
Students already rely on several methods to organize their tasks. The most common are calendar applications (55%) and phone reminders (49.6%). However, a surprisingly high proportion of respondents (40.5%) still depend primarily on their memory.



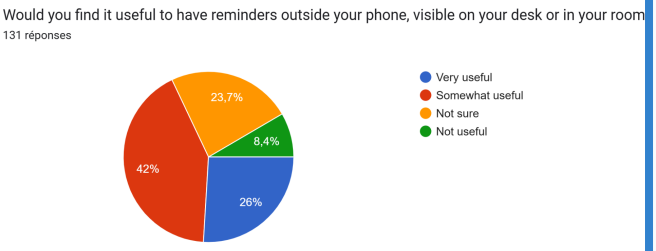
> *Tool Fragmentation: Despite digital options, over 40% of students rely on memory, leading to missed tasks.*

Despite the widespread use of digital tools, the results suggest that these solutions are not always effective. Nearly half of the respondents indicated that phone notifications are only sometimes useful, showing that existing reminder systems often fail to capture attention when needed.

> *Ignored Alerts: Phone notifications are easily dismissed or forgotten, proving ineffective as a sole reminder method.*



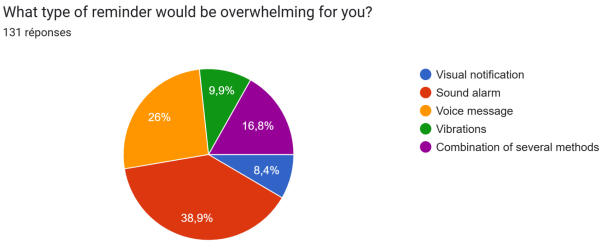
Students also expressed interest in reminders that exist outside the smartphone environment. About two-thirds considered a physical reminder device useful, suggesting that a dedicated object placed in a study space could offer better visibility and reduce dependence on phone notifications.

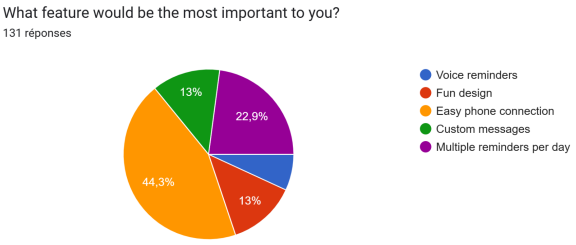


Physical Companion Interest: ~68% of respondents find it useful to have a dedicated physical reminder device on their desk.

When considering reminder modes, students favored non-intrusive alerts. Overwhelming modes like persistent sound alarms or voice messages were seen as undesirable, highlighting the need for gentle, ambient visual cues.

Gentle Reminders: Sound alarms and loud voice prompts are overwhelming; subtle visual alerts are the least overwhelming .





Essential Features: Easy smartphone sync (44.3%) and multi-reminder capability (22.9%) are the top requested capabilities.

Putting these results side by side highlights a clear pattern: phone reminders are used by 49.6% of respondents, yet 52% admit having ignored one and then forgotten the task, and 68% would find a reminder outside their phone useful.

Conclusion

The survey highlights a clear gap between students' organizational needs and the effectiveness of current reminder tools. While most students already use calendars and smartphone reminders, many still forget deadlines and important tasks. The results also show a positive attitude toward a physical reminder device that could remain visible, attract attention more effectively, and

integrate with existing digital tools. These findings support the development of the proposed desk companion as a potential solution to improve task management among students.

Roadmap

The roadmap follows the course agenda and the survey results.
Since 44.3% of students want an easy phone connection, the project is built around a mobile app synced with the device. Because sound alarms were the most overwhelming option, the device uses light and vibration alerts instead.

Software tasks (Git repository, database, app) started first because they do not depend on hardware and can be done remotely. Components are chosen based on what is available at the Fab Lab. Hardware tasks (Wi-Fi connection, display, alerts, 3D printing) are scheduled between 2 and 23 November, when we have access to the Fab Lab.
The work is split by skill: Alexis and Colin handle the code, Chaimae and Diae the 3D design, and the whole team works on the electronics, since it links the app and the enclosure.

This lets the app and the physical device progress in parallel. Finally, the week of 23 November is kept for end-to-end and user testing, which leaves time to fix problems before the A3 poster (7 December) and the Forum (14 December).

#	Task	Who	When	Where	Materials / Tools
1	Conduct the survey (131 responses)	Whole team	21/09 – 05/10 (done)	Online	Google Forms
2	Analyse the survey data and conceptualise the project	Whole team	12/10 – 19/10 (done)	IMT Atlantique	Survey results, Google Docs
3	Project setup: Git repository, Expo project created and cleaned	Alexis, Colin	12/10 (done)	Remote	GitHub, Expo, VS Code
4	Database: schema designed (done), then tables, security rules and test data in Supabase	Alexis, Colin	19/10 – 26/10	Remote	Supabase (PostgreSQL)
5	Choice of functionalities (MVP): create reminders in the app, sync to the device, date/time display, light + vibration alert	Whole team	19/10 – 26/10	IMT Atlantique	Survey results

6	Technology specifications, component list and order	Whole team, Maude	19/10 – 02/11	Remote	ESP32, screen, LEDs, vibration motor, button, USB-C power, cables
7	Device access to data: the ESP32 reads only its own reminders	Alexis, Colin	26/10 – 02/11	Remote	Supabase function, per-device secret key
8	App: sign up, log in, log out	Alexis, Colin	26/10 – 09/11	Remote	Expo, Supabase Auth
9	Choice of the 3D model and enclosure design	Chaimae, Diae	26/10 – 09/11	Remote / Fab Lab	Fusion 360 or Tinkercad
10	Wi-Fi connection, time sync and date display on the device	Whole team	02/11 – 09/11	Fab Lab	ESP32, screen, Arduino IDE
11	App: create, edit and delete devices and reminders	Alexis, Colin	02/11 – 16/11	Remote	Expo, Supabase
12	Device pairing: link an ESP32 to a user	Alexis, Colin	09/11 – 16/11	Fab Lab	Expo, ESP32
13	Reminder alerts on the device: light, vibration, button to mark as done	Whole team	09/11 – 23/11	Fab Lab	ESP32, LEDs, vibration motor
14	3D printing and assembly	Chaimae, Diae	16/11 – 23/11	Fab Lab	3D printer, PLA, screws
15	End-to-end testing and quick user tests with students	Whole team	23/11 – 30/11	IMT Atlantique	Prototype, feedback form
16	Deliverable 2: A3 poster	Chaimae, Diae	30/11 – 07/12	Remote	Canva, PowerPoint or Figma
17	Presentation of the project and poster at the Forum	Whole team	14/12	IMT Atlantique	Poster, working prototype